



Course name: Computer Networks and Data Communication

Code: CSC214

Sheet 1

➤ Part A: Choose the correct answer:

1. What are the five components of a data communications system?
 - A) Sender, Receiver, Message, Medium, Protocol
 - B) Sender, Router, Switch, Medium, Protocol
 - C) Client, Server, Hub, Cable, Protocol
 - D) Sender, Receiver, Packet, Port, Address
2. Which type of data flow allows transmission in both directions simultaneously?
 - A) Simplex
 - B) Half-duplex
 - C) Full-duplex
 - D) Multipoint
3. Which network device operates at Layer 1 (Physical Layer) and broadcasts data to all ports without filtering?
 - A) Router
 - B) Switch
 - C) Hub
 - D) Firewall
4. Which network device operates at Layer 2 (Data Link Layer) and uses a MAC address table to forward data?
 - A) Hub
 - B) Switch
 - C) Repeater
 - D) Router
5. Which device operates at Layer 3 (Network Layer) and routes data between different networks using IP addresses?
 - A) Switch
 - B) Hub
 - C) Router
 - D) Bridge
6. In which network topology do all devices connect to a central managed switch, making it the most common LAN design?



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- A) Bus Topology
 - B) Ring Topology
 - C) Star Topology
 - D) Mesh Topology
7. Which topology provides maximum redundancy and fault tolerance by connecting every device to every other device or multiple others?
- A) Mesh Topology
 - B) Bus Topology
 - C) Star Topology
 - D) Ring Topology
8. What is a key characteristic of Bus Topology?
- A) Centralized switch control
 - B) Devices share a single backbone cable
 - C) Closed circular loop for data flow
 - D) Full duplex communication on every port
9. Which category of network covers a town or a city?
- A) LAN
 - B) MAN
 - C) WAN
 - D) PAN
10. Which firewall type inspects individual packets based on header information like source IP, destination IP, and port numbers using an Access Control List (ACL)?
- A) Packet Filtering Firewall
 - B) Web Application Firewall
 - C) Proxy Firewall
 - D) Stateful Inspection Firewall
11. What is the primary function of a Web Application Firewall (WAF)?
- A) Routing packets between WAN and LAN
 - B) Protecting web servers at the application layer by filtering HTTP/HTTPS traffic
 - C) Repeating electrical signals at Layer 1
 - D) Assigning dynamic IP addresses to hosts
12. Walkie-talkies are a classic real-world example of which data flow mode?
- A) Simplex



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- B) Half-duplex
 - C) Full-duplex
 - D) Multipoint
13. In a point-to-point connection:
- A) More than two devices share a single link
 - B) A dedicated link connects two devices exclusively
 - C) Data is broadcasted to all active nodes
 - D) No protocols are required
14. Which firewall type maintains a conversation list to keep track of active connections and offers more security than basic packet filtering?
- A) Packet Filtering Firewall
 - B) Stateful Inspection Firewall
 - C) Hub-based Firewall
 - D) Layer 1 Firewall
15. What is the primary role of a network protocol in data communications?
- A) To physically amplify electrical signals over long cable distances
 - B) To provide a standardized set of rules that govern how data is formatted, transmitted, and received
 - C) To route packets dynamically between different internet service providers
 - D) To block unauthorized web application traffic and cyber threats

➤ **Part2: Essay Questions**

1. Explain the five fundamental components of a data communications system.
2. Compare and contrast Hubs, Switches, and Routers in computer networks.
3. Describe Star Topology and Mesh Topology, highlighting their main advantages and disadvantages.
4. Categorize networks based on their geographical coverage into LAN, MAN, and WAN, and give an example of each.
5. Compare Packet Filtering Firewalls and Stateful Inspection Firewalls in terms of operation and security.